

BANARAS HINDU UNIVERSITY

B.A./B.Sc. (Hons.) Semester V Examination, 2013-14

Paper No. STA/STB-504: Operations Research

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer any **five** questions. All questions carry equal marks. Symbols have their usual meanings.*

Question 1.

What is meant by a mathematical model of a real situation? Discuss the importance of models in the solution of Operations Research (OR) problems. [14]

Question 2.

- (a) What do you mean by the simplex procedure? Solve the following Linear Programming Problem (LPP) by the simplex method:

$$\begin{aligned} &\text{Maximise } Z = 3x_1 + 5x_2 + 4x_3 \\ &\text{subject to } x_1 + 3x_2 \leq 8 \\ &\quad \quad \quad x_2 + 5x_3 \leq 10 \\ &\quad \quad \quad x_1 + 2x_2 + 4x_3 \leq 15 \\ &\quad \quad \quad x_1, x_2, x_3 \geq 0 \end{aligned}$$

[7]

- (b) Show that the set of all feasible solutions of an LPP is a convex set. [4]

- (c) What do you mean by the two-phase method? Explain how it is used to solve an LPP when an initial basic feasible solution is not obvious. [3]

Question 3.

What do you mean by a transportation problem? Show that the transportation problem can be regarded as a particular case of an LPP. Solve the following transportation problem:

	D_1	D_2	D_3	D_4	Availability
O_1	1	2	3	4	30
O_2	4	3	2	1	20
O_3	2	3	4	5	10
Requirement	6	10	12	15	

[14]

Question 4.

What do you mean by an inventory problem? Describe the Economic Lot Size (ELS) model with uniform rate of demand, infinite rate of production, and no shortages. A manufacturer has to supply his customer with 600 units of his product per year. Shortages are not allowed and the storage cost amounts to Rs. 0.60 per unit per year. The setup cost per run is Rs. 80. Find the optimum run size and the minimum average yearly cost. [14]

Question 5.

What do you mean by duality in LPP? Write the dual of the following LPP and then solve it. Also find the solution of the primal:

$$\begin{array}{ll}\text{Maximise} & Z = 3x_1 - x_2 \\ \text{subject to} & x_1 \leq 4 \\ & x_2 \leq 4 \\ & x_1 + x_2 \leq 5 \\ & -x_1 \leq -1 \\ & x_1, x_2 \geq 0\end{array}$$

[14]

Question 6.

Explain saddle point and zero-sum game. For a 2×2 zero-sum two-person game where the optimal strategies are not pure strategies, show that the optimal mixed strategies for Player A are (x_1^*, x_2^*) and for Player B are (y_1^*, y_2^*) given by:

$$x_1^* = \frac{a_{22} - a_{21}}{a_{11} + a_{22} - a_{12} - a_{21}}, \quad y_1^* = \frac{a_{22} - a_{12}}{a_{11} + a_{22} - a_{12} - a_{21}}.$$

Also find the value of the game to A.

[14]

Question 7. Write short notes on any **three** of the following:

[14]

- (a) Uses of Operations Research
 - (b) Artificial basis technique (Big-M method)
 - (c) Assignment problem
 - (d) Graphical solution of a rectangular game
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